Yixuan Li

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Profile

Senior Data Science student at the Southern University of Science and Technology (SUSTech), with a cumulative GPA of 3.79/4.00 (ranked 8^{th} out of 49; average score: 91.15/100).

Completed the University and Professional Studies (UPS) visiting program at the University of California, San Diego (UCSD), earning a GPA of 4.00/4.00.

Conducting research on uncertainty quantification and interpretability in large language models to build more reliable AI systems, while also exploring AI for Science through machine learning applications in biology.

Research Interests: Uncertainty Quantification, Interpretable Machine Learning, Hallucination Detection, LLM Reasoning, AI for Science (Biology), Medical AI

Education

Southern University of Science and Technology — **Sep. 2022 – Present**
Undergraduate in Data Science — *Shenzhen, China*

- GPA: 3.79/4.00, Weighted Average Score: 91.15/100

University of California, San Diego — **Mar. 2025 – Jun. 2025**
University and Professional Studies (UPS) Program — *San Diego, CA, United States*

- GPA: 4.00/4.00
- Courses: Topics of Data Science: Trustworthy Machine Learning (A+), AI: Search and Reasoning (A+) , Machine Learning Algorithms (A)

Experience

Benchmarking Uncertainty Quantification Methods in Large Language Models — **Feb. 2025 – Present**
Undergraduate Research Assistant — Advisor: Prof. Hongxin Wei — *SUSTech, China*

- Developed a unified benchmarking pipeline for 10+ **semantic uncertainty quantification methods** (e.g., SE, DSE, SAR) on QA datasets using open-source LLMs, evaluating with AUROC, AUARC, and PRR metrics. A GitHub repository is in preparation.
- Conducted an extensive literature review on semantic uncertainty in large language models, intersecting with hallucination detection and confidence calibration, and documented findings through technical blog posts.

Multimodal Learning for Spatial Transcriptomics-to-Proteomics Prediction — **Oct. 2025 – Present**
Undergraduate Research Assistant — Advisor: Prof. Hongxin Wei, Prof.Shimin Shui — *SUSTech, China*

- Developed a **UNet-based model** that reconstructs spatial maps from RNA transcriptomic data to predict protein expression levels at spatial coordinates, achieving an **average Spearman correlation of 0.77** on the validation set.
- Enabled accurate spatial proteomic prediction and provides a foundation for downstream biological analyses. A private GitHub repository will host the implementation (due to participation in an official challenge).

Statistical Analysis of Mental Health Impact among Menopausal Women — **Sep. 2025 – Present**
Undergraduate Research Assistant — Advisor: Prof.Jianqing Shi — *SUSTech, China*

- Performed data cleaning, normalization, and statistical preprocessing on survey datasets of menopausal women in Malaysia.
- Conducted descriptive and inferential statistical analyses on quantitative mental health scores using **R**, applying correlation analysis and hypothesis testing to identify key psychological factors.

Machine Learning for Clinical Decision Support in Orthodontics — **Jun. 2024 – Aug. 2024**
Summer Research Internship — Advisor: Dr. Ying Shan — *Clinical Research Institute, PKUSZ Hospital*

- Performed data preprocessing on clinical craniofacial imaging datasets, dental records, and treatment histories, which involved cleaning, de-duplication, and standardization.
- Developed linear and neural network models based on 16 craniofacial measurement variables to predict patient-specific tooth extraction plans, achieving promising accuracy and internal adoption to support orthodontic treatment planning.

Projects

Concept Bottleneck in Generative Models: Reproduction Study | *Core Member* **Apr. 2025 – Jun. 2025**

- Reproduced the CVPR paper on Post-hoc Concept Bottleneck frameworks (CB-AE, CC) using GAN-based generative models, replicating key quantitative and qualitative results from the original study.
- Conducted additional experiments on concept interpolation, entanglement, and leakage to evaluate model interpretability, robustness, and reliability under concept-level perturbations.
- Delivered a comprehensive reproduction package including Github code repository, presentation slides, and a public technical blog, achieving a course score of **43.4/40 (Rank 1)** under the supervision of Prof. Lily Weng.

Kaggle Competition: Fine-Tuning Gemma-2B for Chinese QA Dataset | *Team Leader* **Oct. 2024 – Jan. 2025**

- Led a team to fine-tune **Gemma-2B** on the GPT-4 Chinese QA dataset. Built a complete fine-tuning pipeline covering data preprocessing, LoRA integration, and evaluation.
- Applied **Retrieval-Augmented Generation (RAG)** for data augmentation, enhancing dataset diversity and contextual grounding, which improved the model's instruction-following ability.
- Evaluated model performance using BLEU, ROUGE, METEOR, and BERTScore metrics, and performed cross-lingual robustness analysis showing stable results across four languages.

Reinforcement Learning on FrozenLake (Course Project) | *Core Member* **Oct. 2024 – Jan. 2025**

- Implemented algorithms (Double DQN, Dueling DQN), trained and analyzed agents on FrozenLake environment.
- Performed hyperparameter tuning and robustness comparison across algorithms, and achieve a performance with 98% accuracy.

Awards

- **First-class Scholarship for Outstanding students, SUSTech** (Top 5%) **2023 – 2024, 2024 – 2025**
- **Third-class Scholarship for Outstanding students, SUSTech** (Top 15%) **2022 – 2023**
- **Second Price** Awarded in the 2024 China Undergraduate Mathematical Contest in Modeling (CUMCM), in Guangdong division **Oct. 2024**
- **Honorable Mention** Awarded in the 2024 Mathematical Contest In Modeling (MCM) **May 2024**
- **Excellent Student Award** Awarded for outstanding students in SUSTech for three times **2022 – 2025**
- **Excellent Volunteer Award** Awarded for outstanding volunteers in SUSTech **2022 – 2023**

Skills

Programming Languages: Python, Java, JavaScript, SQL
Developer Tools: VS Code, IntelliJ IDEA, Google Cloud Platform
Software: Latex, Termius, Canva, Powerpoint, Jupyter Notebook, Git, OmniGraffle, Zotero, Xmind
Language: Mandarin (local), English (fluent: TOEFL 103), Cantonese (fluent)

Leadership / Extracurricular

Media of Shuren College **Sep. 2023 – Aug. 2024**
President *SUSTech*

- Leading a team with 15 Team members. Created a lots of content on college official account and organized activities.
- Award: Excellent College Student Leaders, Third Award of Excellent College Organization.

Pickleball Team **Oct. 2023 – Present**
Member *SUSTech*

- Being a Women's doubles representing member of the school's Pickleball team to compete in the second Guangdong Pickleball Competition, entering the top 16.